

CS 172: Computability and Complexity

Equivalence of CFGs and PDAs & CFL Pumping Lemma

Sanjit A. Seshia
EECS, UC Berkeley

Acknowledgments: L.von Ahn, L. Blum, M. Blum

Formal Definition of **Acceptance**

PDA $P = (Q, \Sigma, \Gamma, \delta, q_0, F)$ accepts a word $w \in \Sigma^*$

where $w = w_1 w_2 w_3 \dots w_m$ with $w_i \in \Sigma_\varepsilon$

if *there exists* a sequence

$(q_0, s_0) \rightarrow (q_1, s_1) \rightarrow (q_2, s_2) \rightarrow \dots \rightarrow (q_m, s_m)$

where

$s_i \in \Gamma^*$ (represent the stack), with $s_0 = \varepsilon$,

$q_m \in F$,

$(q_{i+1}, b) \in \delta(q_i, w_{i+1}, a)$

where $s_i = at$ and $s_{i+1} = bt$, $a, b \in \Gamma_\varepsilon$, $t \in \Gamma^*$

Theorem

Suppose L is generated by a CFG $G = (V, \Sigma, R, S)$

Construct $P = (Q, \Sigma, \Gamma, \delta, q, F)$ that recognizes L

A Language is generated by a CFG

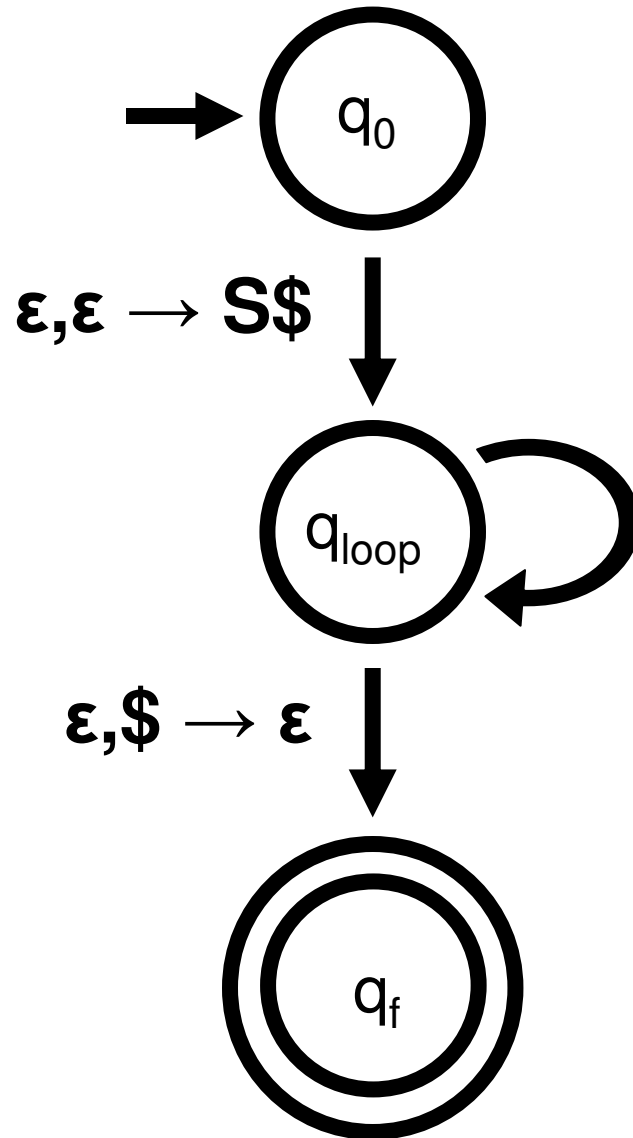


It is recognized by a PDA

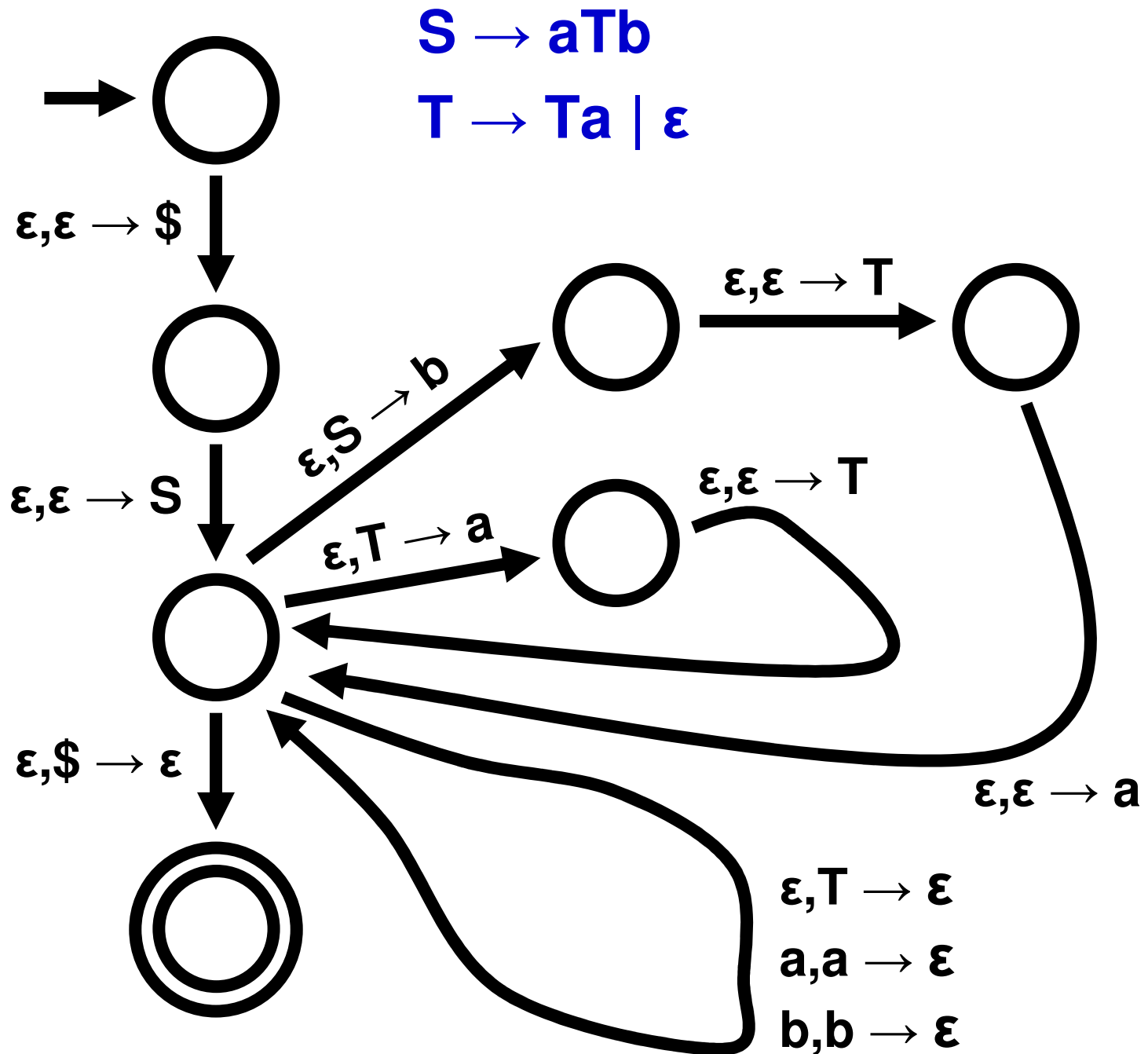
A Language is generated by a CFG

$\Rightarrow$

It is recognized by a PDA



$\epsilon, A \rightarrow w$ for rule $A \rightarrow w$
 $a, a \rightarrow \epsilon$ for terminal a



Theorem

**Suppose PDA $P = (Q, \Sigma, \Gamma, \delta, q, F)$ recognizes L
Construct CFG $G = (V, \Sigma, R, S)$ that generates L**

A Language is generated by a CFG

$\Leftarrow$

It is recognized by a PDA

Proof Ideas

- A_{pq} = variable generating all x that takes P from (p, ε) to (q, ε)
- Formal construction:
 - $V = \{ A_{pq} \mid p, q \in Q \}$
 - $S = A_{q_0 q_f}$
 - Defining R :

Intuition: Derivations correspond to computations of P ;

There are two cases for a derivation from A_{pq}

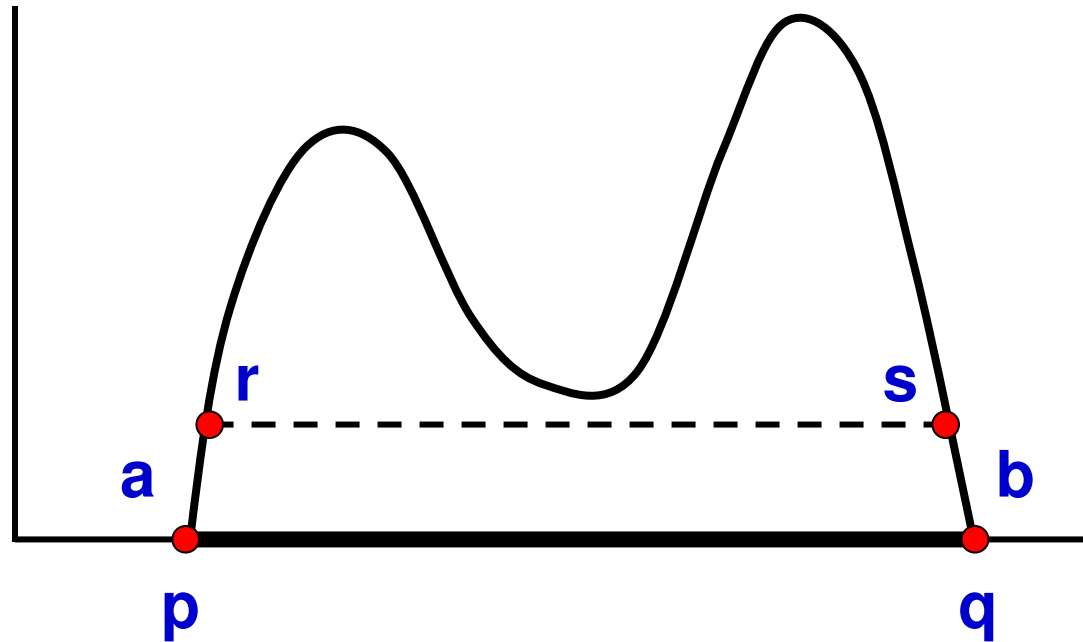
1. Stack is empty only at the beginning and end of the derivation from A_{pq}
2. Stack becomes empty somewhere in between

Case 1

↑
stack height

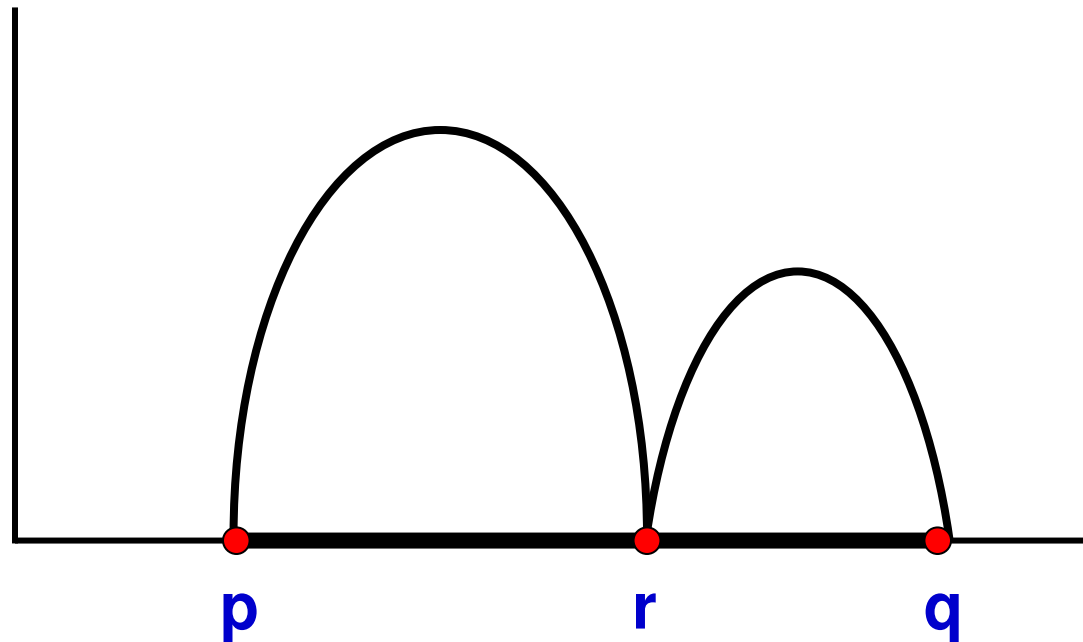
$$A_{pq} \rightarrow aA_{rs}b$$

→
input string



Case 2

$$A_{pq} \rightarrow A_{pr}A_{rq}$$



Recap of Proof Ideas

- A_{pq} = variable generating all x that takes P from (p, ε) to (q, ε)
- Formal construction:
 - $V = \{ A_{pq} \mid p, q \in Q \}$
 - $S = A_{q_0 q_f}$
 - R defined as follows:
 - $A_{pp} \rightarrow \varepsilon \quad \forall p \in Q$
 - $A_{pq} \rightarrow a A_{rs} b \quad \forall p, q, r, s \in Q$
s.t. $(r, t) \in \delta(p, a, \varepsilon)$
 $(q, \varepsilon) \in \delta(s, b, t)$
 - $A_{pq} \rightarrow A_{pr} A_{rq} \quad \forall p, q, r \in Q$

[Proof sketched on whiteboard, see textbook for details]

CFL Pumping Lemma

Let L be a context-free language

Then **there exists p** such that
for all $w \in L$ and $|w| \geq p$

we can write **$w = uvxyz$** , where:

1. $uv^ixy^iz \in L$ for any $i \geq 0$
2. $|vy| > 0$
3. $|vxy| \leq p$

“Surgery” on Parse Trees

Idea: If w is long enough, then any parse tree for w must have a path that contains a variable more than once

